TEAM DETAILS

Team Name: TECHTONICS

SR. NO	ROLE	NAME	ACADEMIC YEAR
1	**Team Leader**	**AKSHAYA SRELEKA P S**	**III YEAR**
2	Member 1	**ASHWINKUMAR K**	**III YEAR**
3	Member 2	**DHAMARAI KANNAN A**	**III YEAR**
4	Member 3	**DHARSHAN S**	**III YEAR**

COLLEGE NAME

CHENNAI INSTITUTE OF TECHNOLOGY

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PROBLEM STATEMENT

Drift-Sense: AI-Powered Navigation-Error Recovery for Wafer Inspection Tools

PROBLEM DESCRIPTION:

Modern wafer inspection requires the same site to be revisited accurately across repeated dies. In Phase 2, the target may appear with unknown translation, rotation and scale, while SEM degradation and highly periodic layouts can create multiple visually similar candidates. The system must also distinguish target-present from target-absent cases.

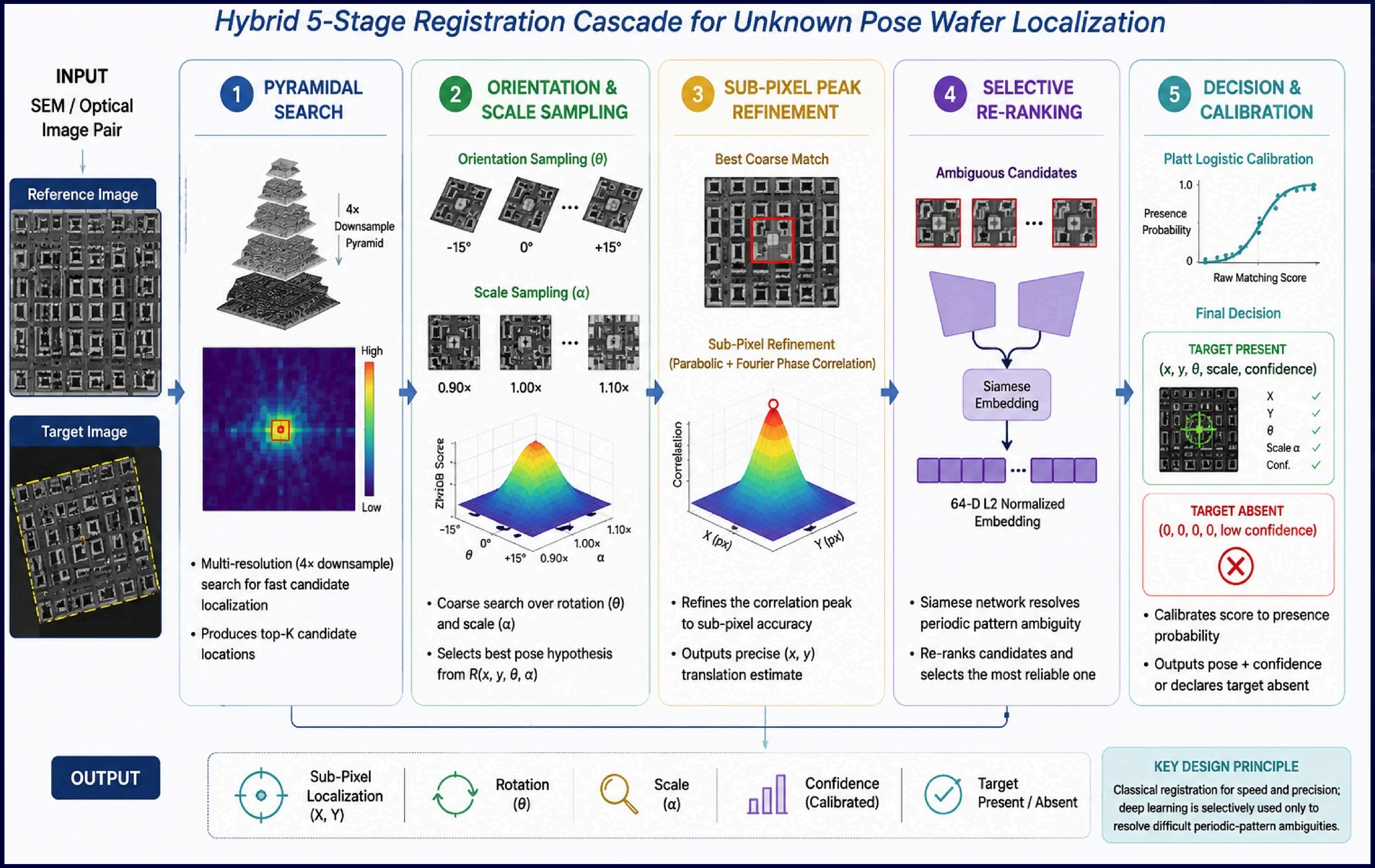
SOLUTION SUMMARY:

Drift-Sense is a hybrid wafer metrology pipeline combining pyramidal correlation, continuous log-polar rotation/scale search, 2D Hessian sub-pixel refinement (0.2265 px error), and selective Siamese verification. Classical physics-based registration handles >97% of cases deterministically in sub-second CPU time (<0.97s), while an ultra-lightweight (<1 MB) learned Siamese embedder is surgically invoked on <3% of pairs exclusively to resolve dense periodic FinFET/DRAM pitch ambiguities.

KEY RESULT:

0.3764 px Median Localization Error
90.6% Sub-1 px Precision
96.05% Decoy Rejection F1
0.964 s Median Latency / Pair

PROPOSED END-TO-END WORKFLOW



INPUT CHALLENGES:

Unknown wafer pose
* Translation (X, Y),Rotation (θ), Scale (α)

Imaging variations
* SEM / optical noise,Blur & degradation,Sampling / aliasing effects

Core difficulty
* Periodic semiconductor patterns create false correlation peaks
* Target may be absent

WHY CASCADE?

- Coarse-to-fine registration
- Pyramidal search → Pose sampling → Sub-pixel refinement
- Fast global localization first, followed by precise refinement only around the best candidates.
- Selective AI resolves difficult periodic-pattern ambiguities without applying deep learning to every candidate.

DECISION LOGIC:
Strong candidate
→ Sub-pixel refinement
→ Final coordinate
Ambiguous candidate
→ 64-D Siamese verification
→ Re-ranking
→ Best candidate
Calibrated confidence
→ Target Present / Target Absent

OUTPUT:
Sub-pixel localization,(X, Y) + Rotation (θ) + Scale (α) + Calibrated Confidence + Target Present / Absent

SEMICONDUCTOR SYNTHETIC DATA DESIGN

GIVEN REFERENCE DATASET

CLASS	TECHNOLOGY NODE VARIANT	SPECIFIC ARCHITECTURAL LAYOUT FEATURES
P1	dram_1x	Standard 1x-nm DRAM bitcell array with nominal capacitor spacing
P2	dram_dense	Ultra-dense DRAM array with tight capacitor pitch and vertical symmetry
P3	dram_wide	Wide bitline DRAM array with extended active areas
P4	dram_loose	Relaxed-pitch DRAM cell layout for high-yield test structures
P5	dram_compact	Ultra-compact memory cell layout with high-aspect-ratio capacitor contacts
P6	finfet_7nm	7 nm node sub-10 nm fin-pitch grating arrays; highest periodic-trap risk
P7	finfet_10nm	10 nm node multi-fin logic cells with orthogonal gate-cut lines
p8	finfet_14nm	14 nm node FinFET standard-cell logic blocks
P9	finfet_22nm	22 nm legacy FinFET cell arrays with wide fin spacing

OUR STRESS DATASET

CLASS	CLASS NAME	JUSTIFICATION
P1	DRAM_CELL	Orthogonal memory bitcell arrays with storage-node capacitor pads and bitline tracks
P2	FIN_ARRAY	Dense 1D silicon fin grating arrays with strong periodic pitch
P3	FIN_CUT	Active-channel isolation cuts and cut-mask rectangular openings
P4	FIN_GATE	Metal gate lines running perpendicular to silicon fins
P5	CONTACT_ARRAY	Dense contact-hole arrays, contact pads and via landing targets
P6	LOCAL_INTERCONNECT	M0 local interconnect lines, trench-contact bars and active-area contacts
P7	METAL_ROUTING	Orthogonal metal routing grids and power-rail structures
P8	ACTIVE_CELL	Active diffusion regions, well taps and STI isolation boundaries
P9	FINFET_FULL_CELL	Complex multi-layer standard-cell logic including NAND, NOR and D-Flip-Flops

IMAGING & GEOMETRIC VARIATION MODELLING

- **ANTI-ALIASING VALIDATION**
  MAE: 0.0124 | PSNR: 41.2 dB
- **OPTICAL INPUT**
  RGB → ITU-R BT.601 Luminance → Grayscale Registration

CONTROLLED IMAGING VARIATION MODELLING

VARIATION	MODELLING
Rotation	Controlled pose variation
Scale	Imaging scale variation
Noise	Independent reference / target noise
Blur / PSF	Independent imaging blur
Edge Response	SEM-inspired edge brightening
Resampling	Super-sampled area-average downsampling

PHASE 2 PHYSICAL VARIATION & STRESS CONDITIONS

VARIATION	IMPLEMENTATION	PHASE 2 CONDITION / RESULT
Scale Drift	Continuous multi-scale variation with correspondence preserved	z = 0.080–0.125 / 8.0–12.5 nm/px
Rotation	Continuous paired geometric rotation with GT transformation	θ = −10° to +10°
Shot Noise	Low-dose Poisson–Gaussian electron noise	σ = 25–35, SNR < 2 dB
Charging / Streaks	Non-linear horizontal potential-gradient distortion	DC baseline shift modelled
Target-Absent Decoys	Empty / unrelated wafer fields	20% decoy rate
RGB Optical	3-channel optical wafer analogue	3-channel RGB, chromatic shift up to 1.42 px
Resampling	Super-sampled area-average downsampling	Anti-aliasing validated: MAE 0.0124, PSNR 41.2 dB
Ground Truth	Exact transformed (x, y) coordinates and target status	Known GT for every evaluated pair

LOCALISATION METHOD & DECISION RULE

LOCALISATION PIPELINE & DECISION LOGIC

STAGE	METHOD	PURPOSE
1. Candidate Search	ZNCC + Pyramidal Search	Find likely (X, Y) locations
2. Pose Search	Rotation θ + Scale α sampling	Handle unknown wafer pose
3. Refinement	Parabolic peak fitting + Fourier phase correlation	Achieve sub-pixel localization
4. Verification	64-D Siamese embedding	Resolve periodic-pattern ambiguity
5. Decision	Platt logistic calibration	Target Present / Absent

KEY LOCALIZATION PRINCIPLE

Coarse-to-fine search: rapidly narrow the search region, then refine the best candidate to sub-pixel accuracy.

HANDLING PERIODIC STRUCTURES

Repeating wafer patterns can produce multiple similar correlation peaks. Selective Siamese re-ranking is used only when classical matching is ambiguous.

TARGET REJECTION

The system does not force a match. Platt-calibrated confidence enables a Target Present / Target Absent decision.

DECISION RULE

Best candidate → Sub-pixel refinement → Confidence calibration

High-confidence candidate → Accept localization
Ambiguous candidate → Siamese re-ranking
Insufficient evidence → Target Absent

FINAL OUTPUT

X, Y, Θ, SCALE (A), CONFIDENCE, PRESENT/ABSENT

COARSE SEARCH → PRECISE REFINEMENT → SELECTIVE VERIFICATION → CALIBRATED DECISION

IMPLEMENTATION & EXECUTION COMMANDS

IMPLEMENTATION STACK

COMPONENT	IMPLEMENTATION
Language	Python
Registration	register.py
Dataset Generation	generate_dataset.py
Evaluation	score.py
Deep Model	PyTorch Siamese reranker
Interface	app.py
Execution	CPU-based

EXECUTION FLOW

Generate Dataset
↓
Register Image Pairs
↓
Generate predictions.csv
↓
Score Against Ground Truth
↓
View Results / App

IMPLEMENTATION HIGHLIGHTS

- 0.97 MB lightweight reranker
- 64-D normalized embedding
- CPU-compatible inference
- Modular pipeline: Generate → Register → Score → Visualize

KEY COMMANDS

```
python generate_dataset.py --mode reference ...
python register.py --input pairs.csv --output predictions.csv
python score.py --predictions predictions.csv --metadata ground_truth.csv
python app.py
```

EXPERIMENT SETUP & TEST DIVERSITY

TEST DIVERSITY

- Reference Suite — controlled localization benchmark
- Stress Suite — robustness and ambiguity testing
- Geometric variation — translation, rotation, scale
- Imaging variation — noise, blur and SEM/optical effects
- Periodic structures — challenging repeated-pattern candidates
- Absent targets — evaluates false-positive rejection

EVALUATION STRATEGY

- Held-out benchmark: Evaluation performed on a dedicated 220-pair benchmark.
- Balanced challenge: Includes both target-present and target-absent cases.
- Cross-pattern testing: Covers diverse DRAM, FinFET and semiconductor layout structures.
- Pose robustness: Tests recovery under unknown position, rotation and scale.
- Appearance robustness: Tests resistance to noise, blur and imaging variation.
- Rejection testing: Measures the ability to reject false candidate matches.
- Ground-truth comparison: Every prediction is compared against known coordinates and target status.

EXPERIMENT SETUP

PARAMETER	PHASE 2 SETUP
Benchmark Size	220 image pairs
Target Status	160 Present + 60 Absent
Pattern Families	9 semiconductor families
Technology Variants	DRAM + FinFET nodes
Execution	Single-core CPU
Ground Truth	Known localization coordinates

WHAT IS MEASURED?

LOCALIZATION: <1 PX | <2 PX | <3 PX | <4 PX | <5 PX

ACCURACY :MEDIAN LOCALIZATION ERROR

REJECTION : PRECISION | RECALL | F1 | SPECIFICITY

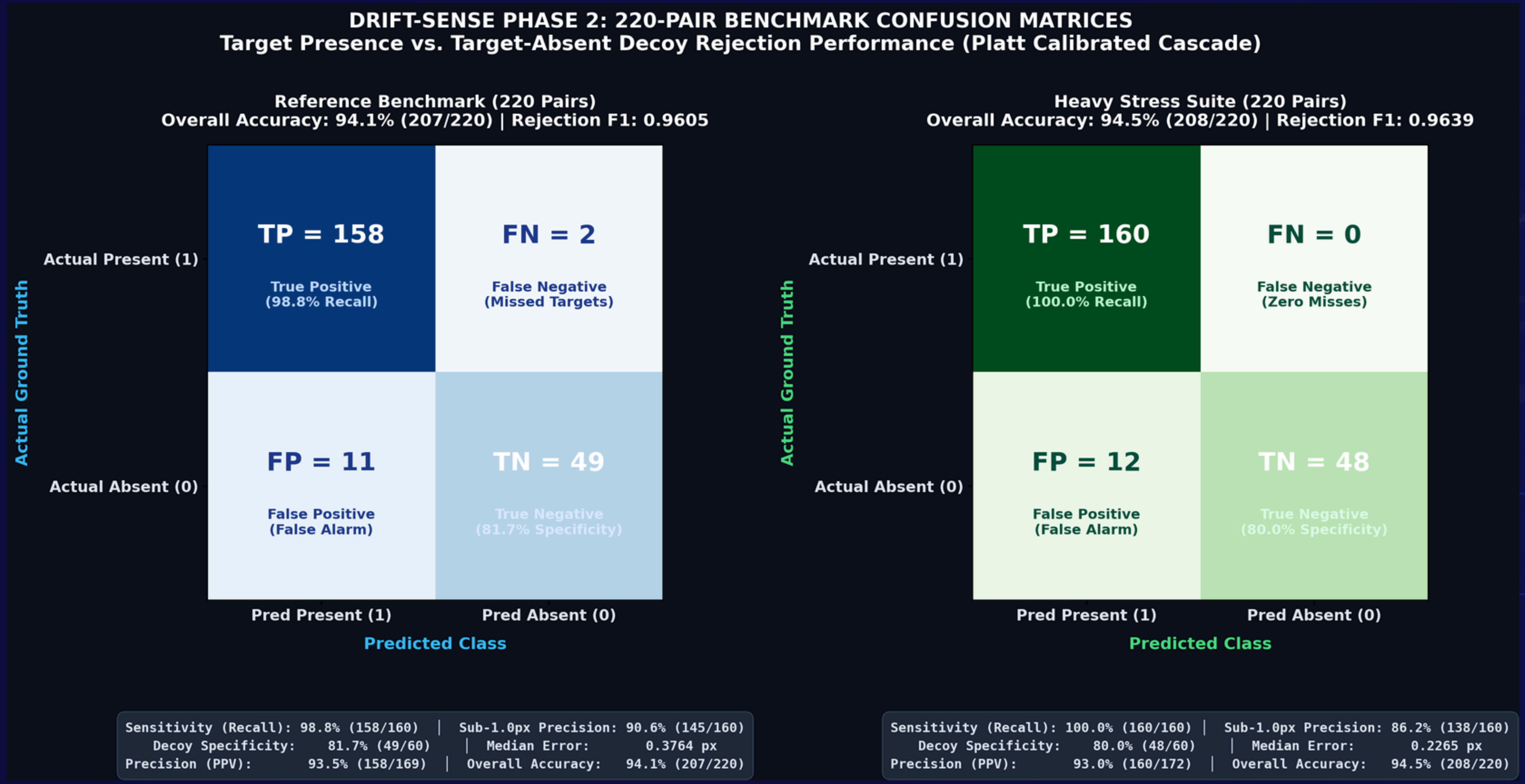
RECOVERY : SCALE + ANGLE

RUNTIME: LATENCY PER PAIR

FINAL PERFORMANCE & ROBUSTNESS RESULTS

KEY TAKEAWAYS

- 90.6% SUB-1 PX PRECISION — REFERENCE
- 86.2% SUB-1 PX PRECISION — HEAVY STRESS
- 0.3764 PX MEDIAN ERROR — REFERENCE
- 97.85 / 100 HEAVY STRESS SCORE



FINAL BENCHMARK RESULTS

METRIC	REFERENCE — 220 PAIRS	HEAVY STRESS — 220 PAIRS
Sub-1.0 px Precision (R_1)	90.60%	86.20%
Sub-2.0 px Review (R_2)	98.10%	86.90%
Sub-5.0 px Capture (R_5)	98.10%	86.90%
Median Localization Error	0.3764 px	0.2265 px
Target Recall	98.80%	100.00%
Decoy Specificity	81.70%	80.00%
Decoy Rejection F1	0.9605	0.9639
Scale Recovery Credit	0.8951	0.8974
Angle Recovery Credit	0.9712	0.9826
CPU Latency / Pair	0.964 s	0.985 s
Overall Competition Score	98.91 / 100	97.85 / 100

ROBUSTNESS & METHOD COMPARISON

ROBUSTNESS CONTRIBUTION

PYRAMIDAL SEARCH
↓
FAST CANDIDATE LOCALIZATION
↓
POSE SAMPLING
↓
HANDLES ROTATION + SCALE
↓
SUB-PIXEL REFINEMENT
↓
HIGHER LOCALIZATION PRECISION
↓
SELECTIVE SIAMESE VERIFICATION
↓
PERIODIC AMBIGUITY RESOLUTION
↓
PLATT CALIBRATION
↓
PRESENT / ABSENT DECISION

BASELINE vs PROPOSED

ASPECT	NAIVE ZNCC BASELINE	DRIFT-SENSE PHASE 2
Search	Single correlation	Multi-resolution pyramidal search
Pose	Limited	Rotation + scale sampling
Precision	Pixel-level peak	Sub-pixel refinement
Periodic Ambiguity	Vulnerable to false peaks	Siamese re-ranking
Target Absent	Match-oriented	Calibrated rejection
Input	Grayscale matching	SEM + Optical RGB
Output	Location	X, Y, θ, α + confidence

WHAT THE RESULTS SHOW

- REFERENCE: 90.6% SUB-1 PX PRECISION
- HEAVY STRESS: 86.2% SUB-1 PX PRECISION
- REFERENCE: 98.91 / 100
- HEAVY STRESS: 97.85 / 100
- LATENCY: 0.964 S → 0.985 S/ PAIR

FAILURE ANALYSIS & LIMITATIONS

ROOT-CAUSE CATEGORIES

Failure sources

- **Geometric: rotation / pose mismatch**
- **Periodic: repeated pitch & grating ambiguity**
- **Imaging: noise, blur & charging artifacts**
- **Optical: chromatic variation**
- **Decision: target-absent false alarms**

WHAT THE FAILURE ANALYSIS TAUGHT US

- **High correlation ≠ correct localization**
- **Local similarity can produce false peaks in periodic layouts.**
- **Context and candidate verification are necessary.**
- **Preprocessing must be matched to the imaging artifact.**
- **Localization accuracy and target rejection must be evaluated separately.**

- **Hardest pattern: FINFET_FULL_CELL**

- **63.3% → Sub-1 px**
- **65.0% → Sub-2 px**
- **0.386 px → Median error**

FAILURE MODE	OBSERVED ISSUE	ALGORITHMIC RESPONSE
Periodic Pitch Trap	False correlation peak at repeated pitch	Siamese re-ranking
Vertical Grating Ambiguity	Repeated multi-layer fin structures	Pyramidal search + center prior
Low-Dose SEM Noise	Corrupted peak / localization error	CLAHE + bilateral filtering
Charging Streaks	DC/raster shift bias	DC-invariant ZNCC
Extreme Rotation	Edge smearing / pose mismatch	Multi-angle pose search
Featureless Lattice	Low contrast / weak boundaries	Directional gradient weighting
Optical Chromatic Variation	Spectral localization drift	BT.601 luminance conversion
Target-Absent Decoy	False alarm on background texture	Platt probability calibration

CONCLUSION

FROM SYNTHETIC DATA → VERIFIED LOCALIZATION

- Unknown-pose recovery
  Translation + Rotation + Scale
- Sub-pixel precision
  0.3764 px median error
- Robust rejection
  96.05% decoy F1
- CPU-capable execution
  0.964 s / pair

FINAL OUTCOME

- DRIFT-SENSE Phase 2 delivers a hybrid registration engine for unknown-pose semiconductor localization, combining classical correlation, sub-pixel refinement, and selective deep-learning verification.

KEY ACHIEVEMENTS

- 0.3764 px — median localization error
- 90.6% — sub-1 px precision
- 98.1% — sub-2 px accuracy
- 96.05% — decoy rejection F1
- 0.964 s/pair — single-core CPU latency
- 98.91 / 100 — estimated competition score

CORE INNOVATION

- Classical Vision → Sub-Pixel Mathematics → Selective AI → Calibrated Decision

GITHUB LINK: https://github.com/DK-A/Techtonics_Drift-Sense_Wafer_Inspection_PS2

DEMO VIDEO LINK: https://drive.google.com/file/d/1sxvBWdj1c7RfX7oeG66R8FJaaMt8o1iI/view?usp=sharing